Nelson Chapter 5 Definitions

thermochemistry the study of the energy changes that accompany physical or chemical changes in matter

energy the ability to do work; SI units joules (J)

work the amount of energy transferred by a force over a distance; SI units joules (J)

potential energy the energy of a body or system due to its position or composition

kinetic energy the energy of an object due to its motion

thermal energy the total quantity of kinetic and potential energy in a substance

heat the transfer of thermal energy from a warm object to a cooler object

temperature a measure of the average kinetic energy of entities in a substance

chemical system a group of reactants and products being studied

surroundings all the matter that is not part of the system

open system a system in which both matter and energy are free to enter and leave the system

closed system a system in which energy can enter and leave the system, but matter cannot

isolated system an ideal system in which neither matter nor energy can move in or out

exothermic releasing energy to the surroundings

endothermic absorbing energy from the surroundings

fusion the process of combining two or more nuclei of low atomic mass to form a heavier, more stable nucleus

fission the process of using a neutron to split a nucleus of high atomic mass into two nuclei with smaller masses

specific heat capacity (*c*) the quantity of thermal energy required to raise the temperature of 1 g of a substance by 1 °C; SI units J/(g·°C)

calorimetry the experimental process of measuring the thermal energy change in a chemical or physical change

calorimeter a device that is used to measure thermal energy changes in a chemical or physical change

enthalpy (*H*) the total amount of thermal energy in a substance

enthalpy change (Δ*H*) the energy released to or absorbed from the surroundings during a chemical or physical change

molar enthalpy change (D*H*r) the enthalpy change associated with a physical, chemical, or nuclear change involving 1 mol of a substance; SI units J/mol

potential energy diagram a graphical representation of the energy transferred during a physical or a chemical change

bond dissociation energy the energy required to break a given chemical bond

standard enthalpy of formation (D*H*f°) the change in enthalpy that accompanies the formation of 1 mol of a compound from its elements in their standard states standard state the most stable form of a substance under standard conditions, 25 °C and 100 kPa

coal slurry a suspension of pulverized coal in water

syngas a synthetic fuel produced from the gasification of coal

efficiency the ratio of the energy output to the energy input of any system